

Proof (sketch): Suppose M_0 is a RM computing χ_{S_0} . From M_0 's program (using the same techniques as for constructing a universal RM) we can construct a RM H to carry out:

let $e = R_1$ and $\ulcorner [a_1, \dots, a_n] \urcorner = R_2$ in

$R_1 ::= \ulcorner (R_1 ::= a_1) ; \dots ; (R_n ::= a_n) ; \text{prog}(e) \urcorner ;$

$R_2 ::= 0 ;$

run M_0

Then by assumption on M_0 , H decides the Halting Problem. **Contradiction.**

So no such M_0 exists, i.e. χ_{S_0} is uncomputable, i.e. S_0 is undecidable.